

Research Methods

Chapter

1

1. [12 marks]

(2013:S3:1)

A psychologist was interested in studying the participation of 17-year-old adolescents in sport. Twenty-five adolescents from two schools participated in the study: 10 from School A and 15 from School B. Participants were volunteers who had responded to advertisements in the schools' newsletters. Each participant completed a survey in which they reported the number of days in the previous week that they participated in any sporting activity. The results were tallied and are shown in Table 1.

Table 1. Frequency of participation in sport by students at School A and School B.

Number of days per week	Number of students from School A	Number of students from School B
0	0	5
1	3	6
2	4	0
3	3	0
4	0	0
5	0	1
6	0	2
7	0	1
Mean	2.00	2.00
Standard deviation	0.82	2.56

(a) Describe briefly the central tendency in the two sets of scores.

[1]

(b) Explain, using the data in Table 1, why it is important to consider a measure of dispersion as well as a measure of central tendency in scores.

[2]

- (c) (i) Explain the difference between the terms 'population' and 'sample' in psychological research. [2]

- (ii) State whether or not the results of the study described above are likely to represent the population of 17-year-old adolescents. Outline **one** reason for your answer. [2]

- (iii) Outline **one** way in which researchers can ensure that their samples are representative of the population. [1]

- (d) Identify whether the study described above is an experimental or non-experimental study. Explain briefly the reason for your answer. [2]

- (e) (i) Give **one** limitation to the reliability of the data collected in this study. [1]

- (ii) State **one** way in which the researchers could increase the reliability of the data collected in the study. [1]

2. [8 marks]

(2013:S3:2)

A psychologist wants to investigate students' attitudes toward fast food advertising on television.

- (a) (i) Name **two** qualitative methods the psychologist could use to collect data. [2]

- (ii) Name **one** quantitative method the psychologist could use to collect data. [1]

- (b) (i) Outline **two** benefits of using qualitative methods for collecting research data. [2]

- (ii) Outline **two** limitations of using qualitative methods for collecting research data. [2]

- (c) State **one** way in which the role of the experimenter is different in qualitative research compared with quantitative research. [1]

3. [20 marks]

A psychologist was interested in studying the characteristics of young people who do volunteer work.

- (a) Name **two** non-experimental (descriptive) research methods that the psychologist could use. [2]

- (b) Name **two** qualitative methods that the psychologist could use. [2]

- (c) The psychologist designs a subjective quantitative measure to collect information about the personality characteristics of volunteers, using a Likert scale.

- (i) Name each of the **five** points on the line below to show a Likert scale. [1]

|-----|-----|-----|-----|

- (ii) Describe **one** advantage of using a Likert scale compared with a checklist. [2]

The psychologist was interested to find out whether there is a relationship between the number of hours spent volunteering and self-esteem in young people.

- (d) Is this an experimental or correlational study? Explain the reason for your response. [2]

The psychologist collected data on the self-esteem of ten young people who were currently involved in volunteer work in their community, using the Likert scale.

The Likert scale produced scores for each person ranging from 10 to 50. Higher scores indicated higher levels of self-esteem.

The self-esteem scores for the 10 young people were:

30, 32, 38, 40, 45, 43, 46, 38, 45, 46

- (e) (i) Complete the frequency table below.

[1]

Self-esteem score	Number of people
11–15	
16–20	
21–25	
26–30	
31–35	
36–40	
41–45	
46–50	
Total	

- (ii) The psychologist is writing a report on the results of the research and needs to include a graph of the frequency of self-esteem scores. Graph the results from the frequency table in part (e)(i) on the axes below. [6]

Frequency of self-esteem scores in volunteers



The psychologist then collected data from a sample of ten young people who had never done any volunteer work.

The psychologist calculated the mean self-esteem score for the volunteer group and the non-volunteer group.

	Mean self-esteem score
Volunteer group	40.3
Non-volunteer group	38.5

A statistical test of the difference between the two means showed $p > 0.05$.

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- (f) (i) State whether the difference between the two means was statistically significant. [1]

- (ii) Describe the meaning of 'statistically significant'. [2]

- (iii) Outline the conclusion that the researcher should draw from the statistical results. [1]

4. [20 marks]

(2014:S3:1)

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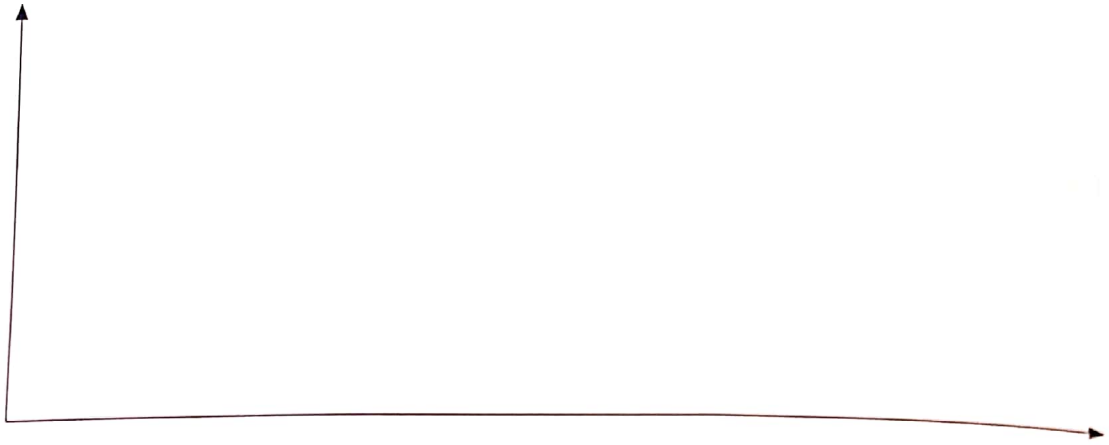
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5. [14 marks]

(2014:S3:2)

Professor Zheng of Smart University wanted to test whether a vitamin supplement helped to improve students' grades. To test her idea, she told her class she was giving them all a supplement that would make them smarter. She told them they all had to participate.

Professor Zheng divided the class into two equal groups. Group A received the vitamin supplement and Group B received an inactive substance (sugar pill).

Over the two week trial, Professor Zheng spent a lot of time with the students gathering data. She was excited to find that the results showed that all students' grades had improved. She concluded that the vitamin supplement had worked.

- (a) (i) Name **three** rights that people have as participants in research. [3]

- (ii) Outline **two** pieces of information about a research study that a researcher must provide to participants so that they can give informed consent. [2]

- (b) (i) Identify the term that refers to a substance with no known medical effects used as a control in an experiment. [1]

- (ii) Explain why Professor Zheng gave one group a substance with no known medical effects. [1]

- (c) Explain what Professor Zheng should do after the experiment has ended to make sure the research has been conducted ethically. [1]

- (d) Outline how Professor Zheng may have affected the results of the experiment. [1]

- (e) (i) Outline **three** ethical issues associated with this research. [3]

- (ii) Identify **two** sources of error in the design of this experiment. [2]

6. [14 marks]

(2015:S3:1)

Researchers were interested in finding out how much time students at a small rural high school spent each day on using electronic devices for recreation. They surveyed all the students at Ridge High School and collected the data shown in Figure 1.

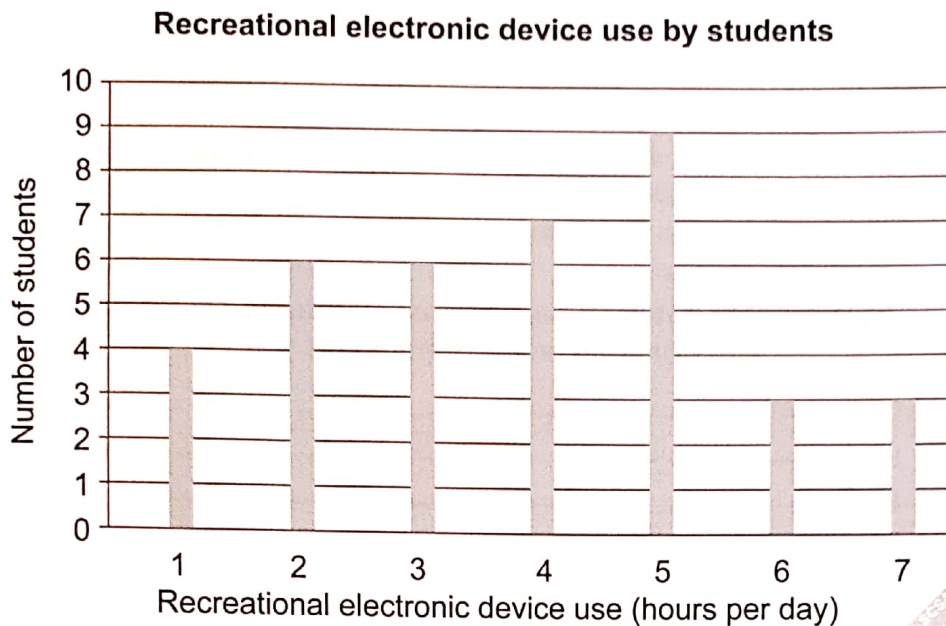


Figure 1

Measures of central tendency for the data in Figure 1 are as follows:

- mean = 3.84
- median = 4.

(a) Identify the mode of the data in Figure 1. [1]

(b) A new student starts at Ridge High School. This student does not own any electronic devices and therefore spends 0 hours per day using electronic devices for recreation. State the effect on each measure of central tendency if the score of 0 was added to the data in Figure 1.

(i) Effect on mean: [1]

(ii) Effect on median: [1]

(iii) Effect on mode: [1]

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- (c) The American Academy of Paediatrics recommends that young people spend no more than 2 hours per day using electronic devices for recreation. Outline what the data in Figure 1 indicate about the use of electronic devices by the students at Ridge High School. [2]

Teachers at Ridge High School wanted to further investigate the use of electronic devices for recreation by their students by examining the relationship between the amount of use and students' achievement test scores. The data is shown in Figure 2.

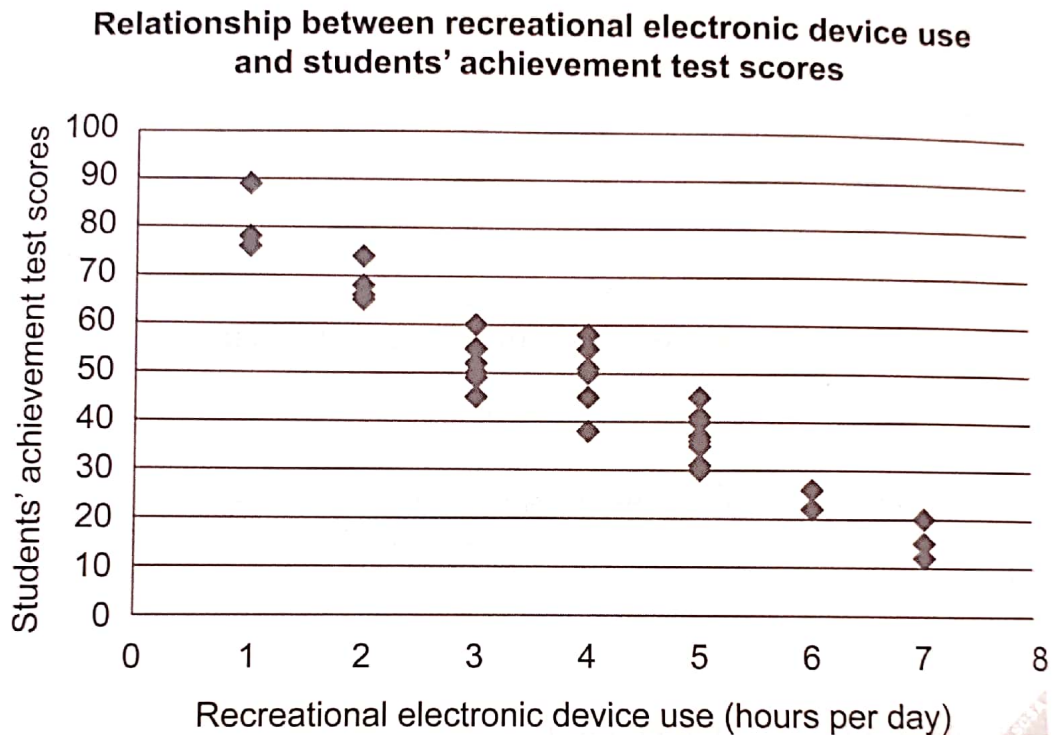


Figure 2

- (d) (i) Name the type of graph shown in Figure 2. [1]
- (ii) Identify the type of relationship between the variables shown in Figure 2. [2]
- (e) The researchers calculated the correlation between the variables in Figure 2 and found $p < .05$. State whether this result was statistically significant. [1]

- (f) Describe what it means when psychologists find a 'statistically significant' result from a statistical test. [2]

- (g) The researchers concluded that using electronic devices for recreation for more than 2 hours per day causes lower achievement test scores in students. State whether this was a correct conclusion to draw and explain your response. [2]

7. [12 marks]

(2015:S3:2)

Researchers at Highpoint University were interested in finding out whether there was any difference in the likelihood of seeking help from psychologists between people who live in rural and remote areas and people who live in urban areas. They surveyed 200 people (100 rural and remote and 100 urban) on how likely they were to seek psychological help if they needed it.

- (a) (i) Identify the independent variable for the study. [1]

- (ii) Identify the dependent variable for the study. [1]

- (b) Write an operational hypothesis for this study. [2]

- (c) Identify **two** variables the researchers would need to control (i.e. take account of) when conducting this study. [2]

One:

Two:

The researchers collected data using the 'Attitudes towards Psychologists Questionnaire' to determine how likely a person was to seek help from a psychologist.

(d) (i) State what psychologists mean when they say that a questionnaire is 'reliable'. [1]

(ii) Name and describe **one** type of reliability a psychologist could test for the questionnaire. [2]

(e) (i) State what psychologists mean when they say that a questionnaire is 'valid'. [1]

(ii) Name and describe **one** type of validity a psychologist could test for the questionnaire. [2]

8. [9 marks]

(2015:S3:3)

(a) Identify **two** features that make psychological research 'scientific'. [2]

One: _____

Two: _____

- (b) In 1963, Stanley Milgram conducted his well-known obedience experiment in which he had participants believe they were giving a fellow participant (who was actually an associate of Milgram) increasingly strong electric shocks. Describe briefly the most significant ethical issue associated with this study and explain why it was a problem. [3]

- (c) Describe **two** ways in which researchers should behave professionally with participants when they are conducting psychological research. [4]

One: _____

Two: _____
